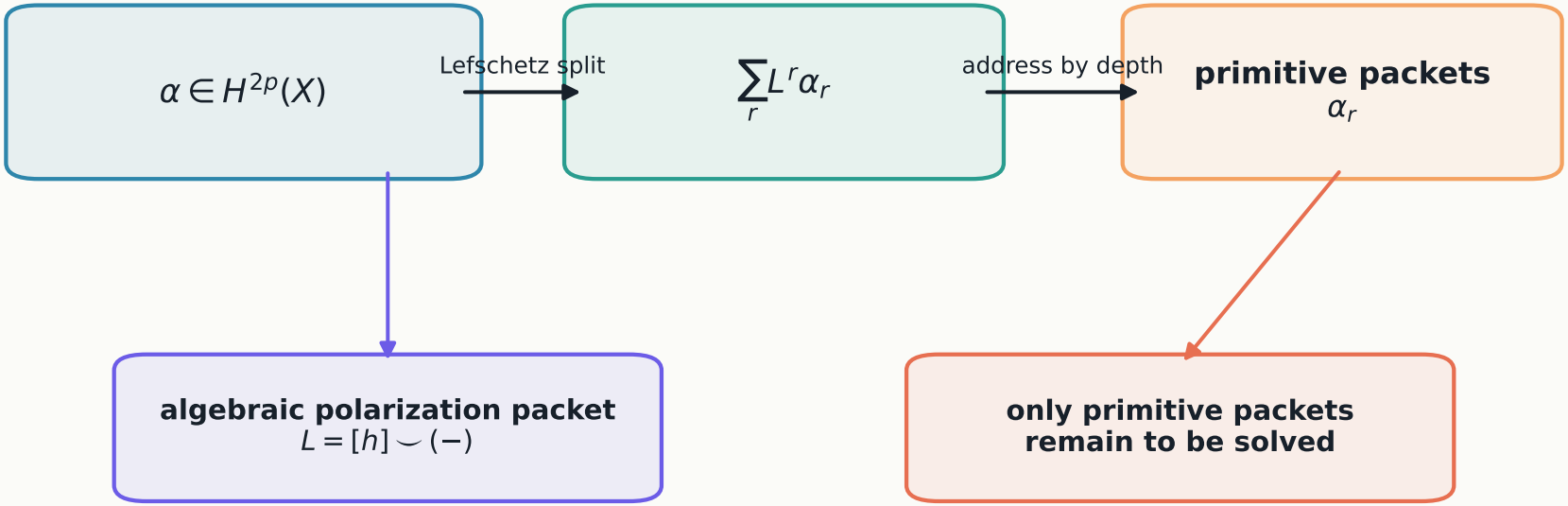Lefschetz decomposition removes algebraically generated descendants